Lecture plan · for the lecturer

Teaching programming and algorithms at university: unit check

Programming · University · Unit check

Time	Materials
40 min	Projector or board, and the lecture slides.

Learning goals

- Asymptotic analysis
- Recursion and induction
- Notional machines
- Assessment at scale

Activity: Peer-instruction review

1. Vote (10 min): Show each question; students answer alone and commit to an answer.
2. Discuss (15 min): Where answers split, pairs argue for two minutes and vote again.
3. Explain (10 min): Work through the questions most students missed, showing the reasoning step by step.
4. Plan (5 min): Note which misconception to revisit next week and how.

Discussion

- Which misconception will you target first?
- Which lab will you add?
- How will you change your assignments?

Exit ticket

- One change to my programming course this term, and how I will know it worked.

Try it in your next lecture
Bring one redesigned assignment, its hidden tests and your AI policy to your next department meeting for review.

Seminar sheet · for students

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Name _____ Class _____ Date _____

1 **$5n^3 + 2n$ is**

◯ $\Theta(n^3)$ ◯ $\Theta(n)$ ◯ $\Theta(n^5)$

2 **Solve the recurrence.**

$T(n) = 2T(n/2) + 1$ gives $\Theta($ _____ $)$.

Word bank: n · log n · n log n

3 **Big-O gives a lower bound on running time.**

◯ True ◯ False

4 **fact(n) with no reachable base case ends in**

◯ A stack overflow ◯ Returning 0 ◯ A syntax error

5 **Order the induction proof.**

☐ Conclusion

☐ Inductive step

☐ Inductive hypothesis

☐ Base case